

CSE200: Computer Aided Engineering Drawing

Tahsin Tariq Banna

Lecturer,
Department of Computer Science and Engineering,
East West University

What was it like before?



General Motors Technical Center in Warren Michigan.

What was it like before?



Drafting department in the late 1970s



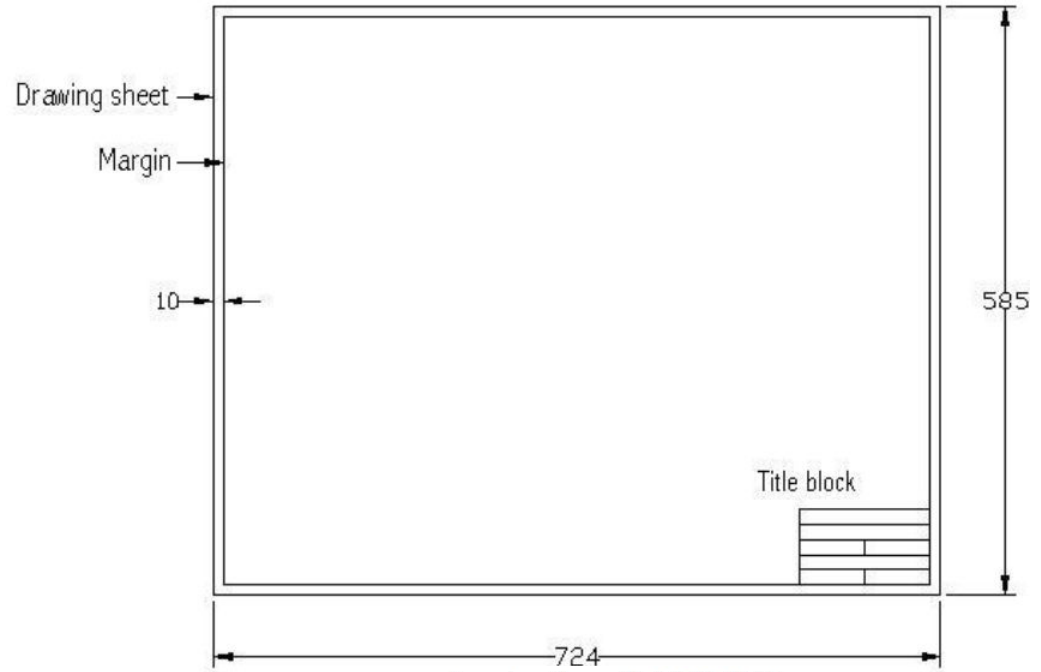
Urban designers before AutoCad

Here draftsmen are correcting geological survey printing plates, at The Department of Interior Geological Survey Section, Washington DC. 1939.



Necessary Instruments

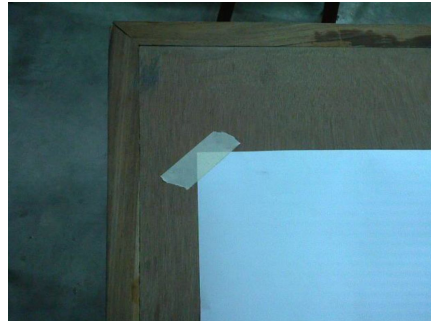
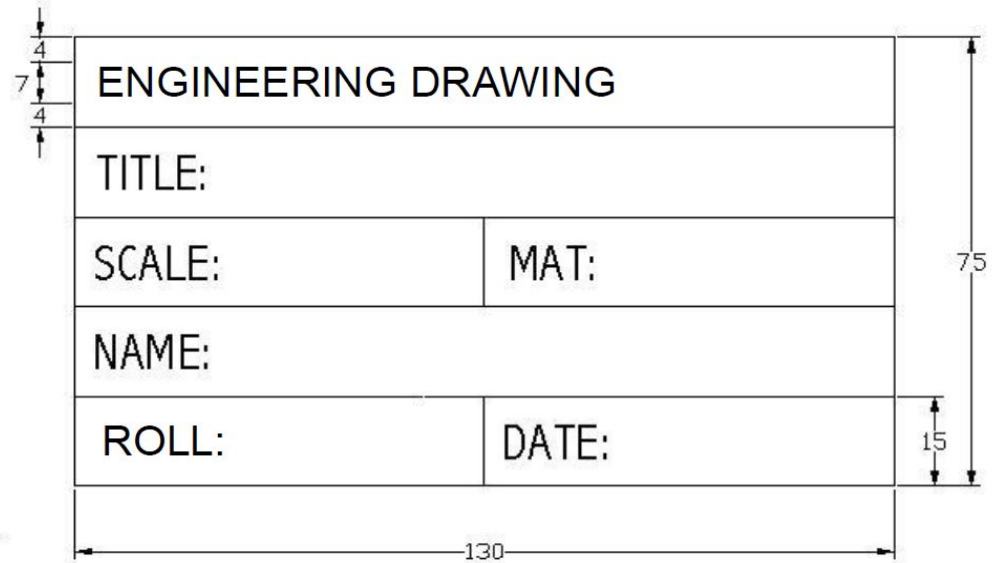
- Drawing Board
- Drawing sheet
- Drafting tape
- T-square square
- Set square
- Compass and divider
- Protractor Protractor
- Pencil (HB+2B)
- Eraser
- Sharpener



<i>Drawing Sheet (IS 10711:2001)</i>	
<i>Designation</i>	<i>Size (mm)</i>
	<i>Length × Width</i>
A0	1189 × 841
A1	841 × 594
A2	594 × 420
A3	420 × 297
A4	297 × 210

Necessary Instruments

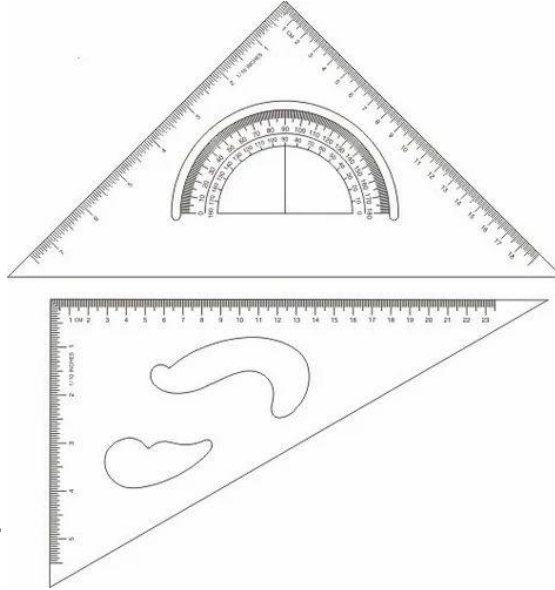
- Drawing Board
- Drawing sheet
- Drafting tape
- T-square square
- Set square
- Compass and divider
- Protractor
- Pencil (HB+2B)
- Eraser
- Sharpener



Necessary Instruments



T Square/T Scale

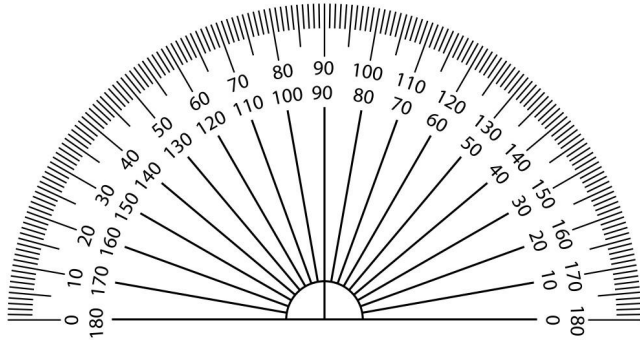


Set Square

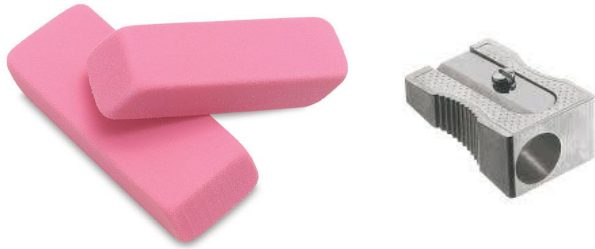


Compass and divider

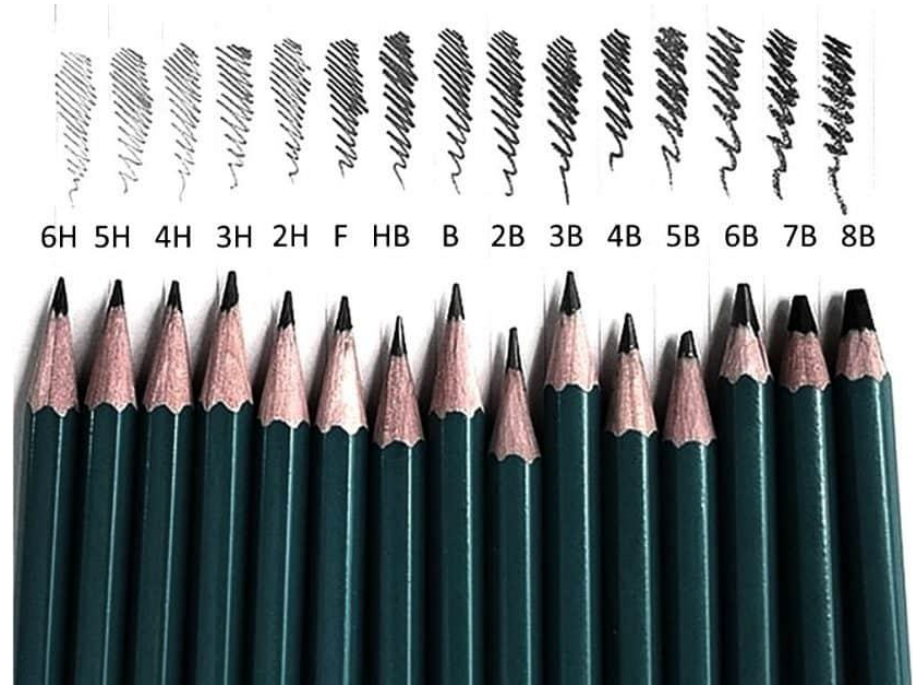
Necessary Instruments



Protractor



Eraser and Sharpener



Pencils

Lettering

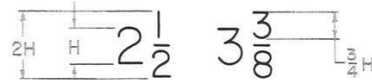
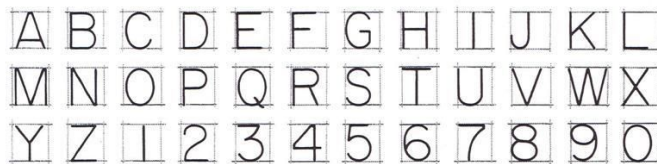
Letters are written in a single stroke, either horizontally or vertically. A single typography is maintained throughout the drawing.

Stroke, kerning, spacing, etc might vary by typographical fonts.

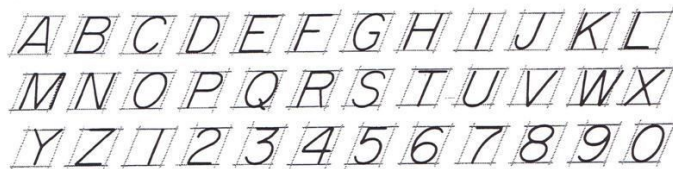
Lettering



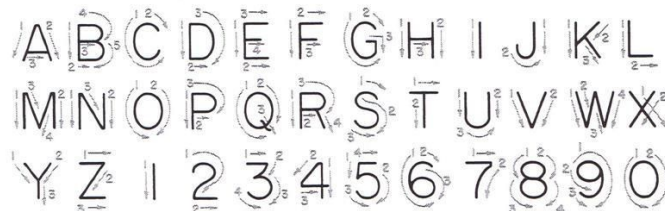
Single-Stroke Gothic Alphabet



VERTICAL SINGLE STROKE GOTHIC ALPHABET



INCLINED SINGLE STROKE GOTHIC ALPHABET



RECOMMENDED SEQUENCE FOR MAKING SINGLE STROKE GOTHIC ALPHABET

Copyright
The Goodheart-Willcox Company, Inc.

RM 7-1

Unit 7 Lettering 101

Line Thickness

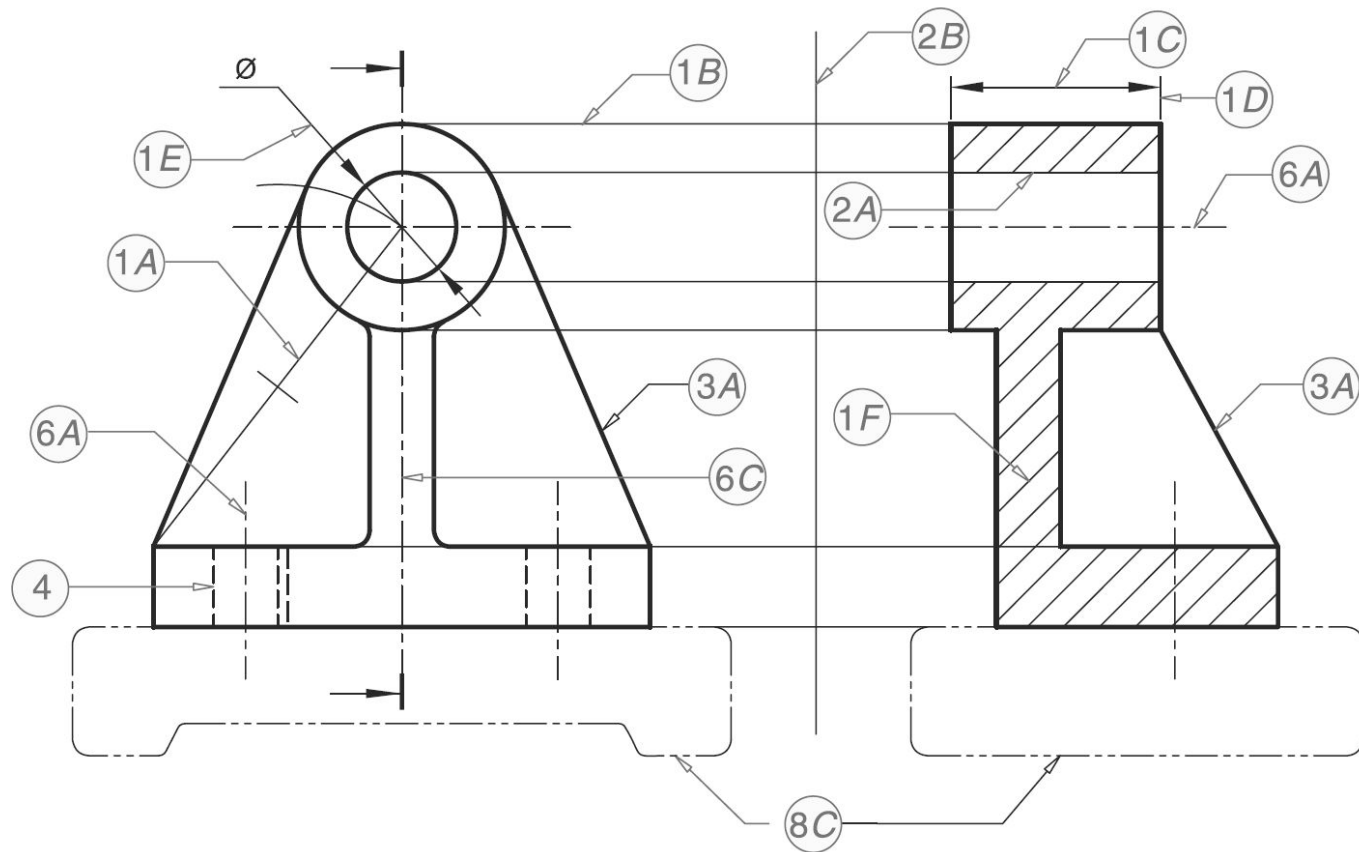
Table 2.1 Types of Lines

S. No.	Line Type	Representation	Width*	Applications
1	Continuous NARROW		THIN (0.13 mm)	A. Construction lines, B. Projection lines, C. Dimension lines, D. Extension lines, E. Leader lines, F. Section (Hatching) lines, G. Outlines of revolved sections, H. Imaginary lines of intersection
2	Continuous WIDE		MEDIUM (0.25 mm)	A. Visible outline of sectioned surface, B. Reference lines in projection, C. Ground Lines
3	Continuous EXTRA-WIDE		THICK (0.5 mm)	A. Visible outlines, B. Lines of special importance
4	Dashed NARROW		THIN	Hidden lines or edges
5	Dashed WIDE		MEDIUM	Hidden lines or edges
6	Long Dashed Dotted NARROW (Chain NARROW)		THIN	A. Centerlines, B. Lines of symmetry, C. Cutting planes
7	Long Dashed Dotted WIDE (Chain WIDE)		MEDIUM	Cutting planes (at ends & change of direction)
8	Long Dashed Double Dotted NARROW (Phantom NARROW)		THIN	A. Locus lines, B. Alternative and extreme positions of movable parts, C. Outlines of adjacent parts
9	Continuous Freehand NARROW		THIN	Short break line
10	Continuous Zigzag NARROW		THIN	A. Long break line, B. Limits of partial or interrupted views

The NARROW, WIDE & EXTRA-WIDE lines are respectively referred as THIN, MEDIUM & THICK lines in this book

*Based on line group 0.25. See Table 2.2.

Line Thickness



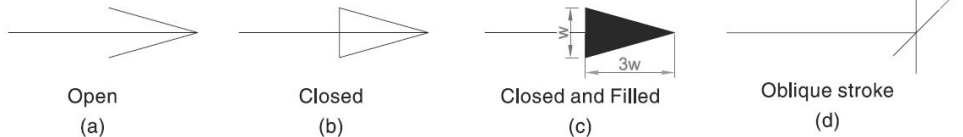
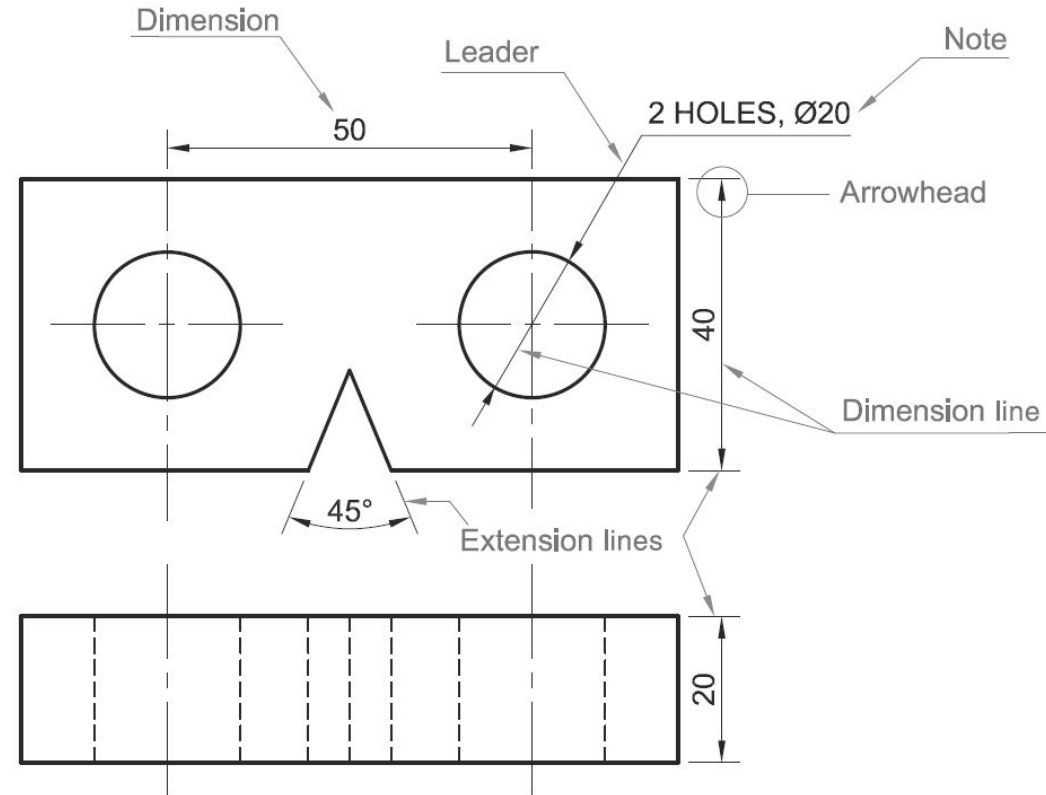
Dimensioning

Dimensioning refers to the act of giving dimensions, i.e., length, width, height, diameter, etc., of the object.

Formally, a dimension as a numerical value expressed in appropriate units of measurement and indicated graphically on technical drawings with lines, symbols and notes.

Dimensioning is done using the following:

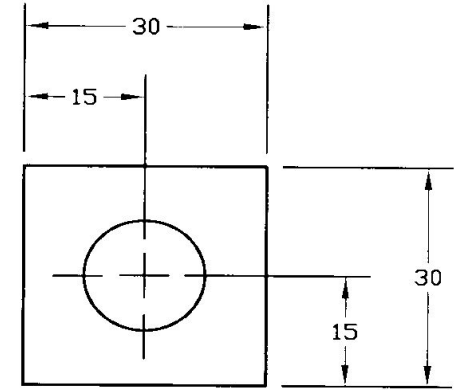
- Arrowheads
- Extension Line
- Dimension Line
- Leader Line



Dimensioning

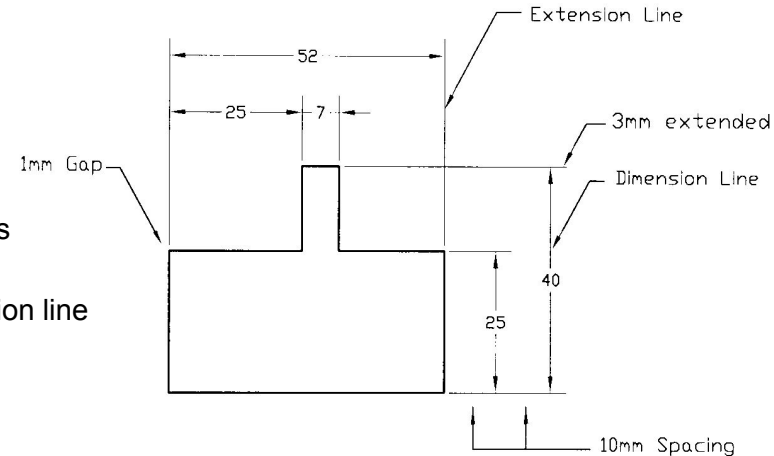
EXTENSION LINE

- A gap of 1 mm has to be kept in between the extension line and the visible line.
- An extension line should extend about 3 mm from the outermost dimension line.
- Extension lines may cross each other without a break.
- Centerlines may be used as extension lines.
- Extension lines are usually drawn perpendicular to dimension lines.



DIMENSION LINE

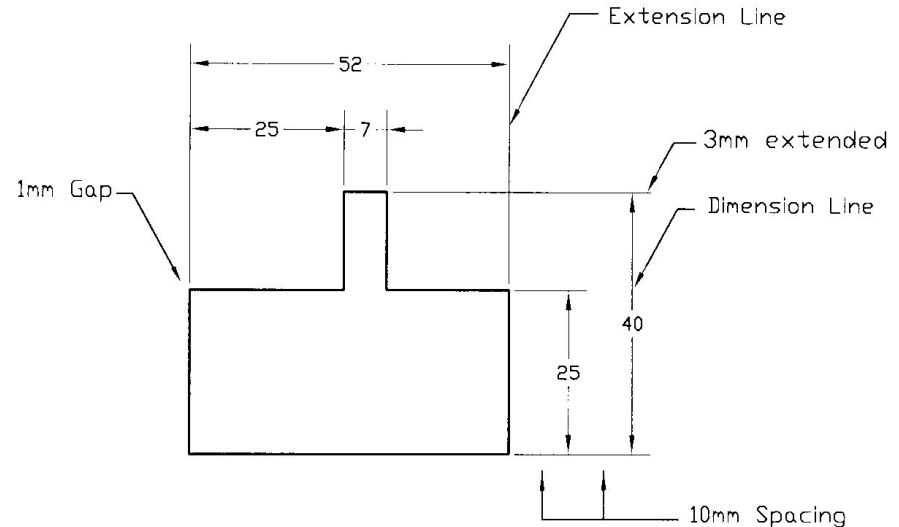
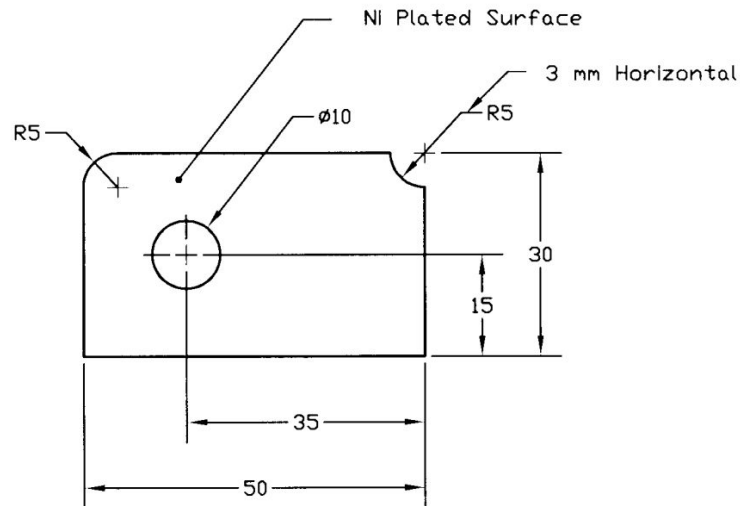
- Dimension line should be approximately 10 mm away from the visible line.
- The spacing between the consecutive parallel dimension lines may also be considered as 10 mm.
- Dimension lines should usually be placed outside the view unless it becomes necessary.
- When the space in between the extension lines is too small to insert dimension line completed with arrowhead, it may be provided outside the extension line.



Dimensioning

LEADERS

- Should always be inclined at an angle of 60° preferably and 45° occasionally with a 3 mm horizontal bar.
- Is either terminated by an arrowhead on a line or a small dot.
- Should not cross each other.
- To dimension a circle or an arc the leader should be so drawn that if it is imagined to extend it must pass through the center of the circle or the arc.
- All notes and dimensions in a leader have to be provided in the horizontal direction.



Softwares



Thank you